

(Non) Pharmacological Modalities in Acute Procedural Pain

To the Editor:

We have read with interest the randomized nonblinded trial by Kristensen et al.¹ The authors studied the contribution of hospital clowns (HCs) on the experience of pain in children aged 4 to 15 years following an acute procedural pain specifically venipuncture. Assessing pain following venipuncture using self-reported pain scales, the study revealed a pain-relieving effect for children older than 6 years. In this study, the authors highlighted the value of one particular non-pharmacological modality for pediatric pain relief and should be congratulated for that. However, we believe there are some important caveats to highlight when interpreting and applying study results.

Overall, needle-related procedures such as venipuncture are recognized as the least well-controlled and most distressing pain and fear-inducing procedures for children in hospitals.² Lack of appropriate pain and fear control will eventually lead to adverse effects on both short-term and long-term psychological health and on a child's compliance with further procedures.²

Nonpharmacological modalities such as distraction provided by HCs therefore indeed are important to consider to help downregulate nociception and pain-related fear, and there is increasing evidence in favor of using such strategies as an inseparable part of any procedural care.³ However, the clinical context in which acute procedural pain occurs is critical to consider when implementing pain treatment. Depending upon the complexity of the clinical scenario, different combinations of approaches may be more suitable, for example, other child's appropriate audio-visual distraction techniques⁴ than HCs, application of local anesthetics, nitrous oxide, ketamine, or even general anesthesia when indicated. Overall, clinicians should be encouraged to reduce both procedural pain and fear with a multimodal and flexible approach that is not

constrained by the limits of any one particular intervention.

In the study by Kristensen et al,¹ conclusions about the HC intervention also were based on pain self-report obtained a few minutes after the venipuncture procedure. Self-reporting typically is regarded as the best method for measuring pain in verbal children. However, some young children may have difficulty accurately reporting pain in this way particularly if asked to do so several minutes after a painful event has occurred. Scores on the primary outcome measure on which study conclusions were based therefore may have been downwardly biased. The authors ideally should have combined the self-reporting pain scores with observational scales at the time of procedure and/or included parent proxy rating as recommended.⁵

In conclusion, implementation of HCs in acute procedural pain may be useful in certain clinical situations with some children. Regardless of the availability of HCs at any given site, however, clinicians should be encouraged to always provide children with the benefit of combined use of pharmacological and nonpharmacological strategies.

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child, parent, nurse, and independent observer. *Paediatr Anaesth*. 2014;24:1127–1131.

Transversus Thoracic Muscle Plane Block for Breast Surgery

To the Editor:

We read with interest the article by Wang et al¹ on The efficacy of ultrasound-guided type II pectoral nerve blocks (PECS) in perioperative pain management for immediate reconstruction after modified radical mastectomy and we congratulate the authors for the elegant study design and for the excellent clinical outcome for the patients.

PECS have been used in some clinical setting as regional anesthetic technique for breast and thoracic surgery as it can anesthetize the lateral mammary region, the intercostobrachial and lateral cutaneous branches of the intercostal nerves (T2–T6), the medial cutaneous nerve of the arm and forearm, and the long thoracic and thoracodorsal nerves.²

PECS have also been used to treat chronic pain after breast and thoracic surgery with interesting results.^{3,4}

The major problem with PECS in breast surgery is that it cannot completely block the internal mammary region. Thus an intravenous analgesia is required and has been correctly used by Wang et al¹ in the present study who administered fentanyl during the intervention and morphine to prevent postoperative pain.²

The thoracic transversus muscle plane block (TTP) is a recently described fascial plane block used to anesthetize the anterior branches of the intercostal nerves from T2–T6 and the efficacy has been demonstrated for breast surgery and for median sternotomy.^{2,5}

We think that TTP could have been a useful adjuvant to PECS to improve the quality of regional anesthesia provided by Wang and colleagues, we also think that TTP should